

Multiple Choice (10 marks)

1. Which of the following is not a function of vitamin D?
 - (a) Lipid-soluble antioxidant in cell membranes
 - (b) Fast responses to increase calcium absorption with no change in gene expression
 - (c) Regulation of differentiation of adipocytes
 - (d) Regulation of bone turnover
2. The respiration quotient (RQ):
 - (a) is defined as the ratio of VO_2 divided by VCO_2 .
 - (b) decreases on a high carbohydrate diet.
 - (c) increases with fasting.
 - (d) goes beyond a value of 1.0 when exogenous carbohydrate is converted to endogenous fat.
3. Which of the following is a function of vitamin B₆?
 - (a) Synthesis of TMP (thymidine monophosphate)
 - (b) Decarboxylation of amino acids to form amine neurotransmitters
 - (c) Synthesis of methionine from homocysteine
 - (d) Carboxylation of pyruvate to oxaloacetate
4. In which form is selenium found in the 25 human selenoproteins?
 - (a) Selenophosphate
 - (b) Selenocysteine
 - (c) Selenohistidine
 - (d) Selenate
5. What mechanisms potentially accelerate age related cognitive decline (as of 2020 medical knowledge)
 - (a) Glucose neurotoxicity
 - (b) Insulin insensitivity
 - (c) Systemic inflammation
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: Obese individuals have a higher basal metabolic rate primarily because they have more adipose tissue than lean individuals.
7. True or False: Biotin plays a crucial role in the carboxylation of pyruvate to oxaloacetate in metabolic processes.
8. True or False: Underwater weighing is the most direct method for measuring body composition.
9. True or False: If survival time with a disease increases, the prevalence of that disease in the population will also increase.
10. True or False: Prenatal programming only impacts children who are born with low birth weight.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of proteins in the absorption of cholesterol in the small intestine, and analyze why the LDL receptor is not involved in this process.
12. Discuss the implications of osteopenia and osteoporosis diagnoses in women over age 80, particularly focusing on how their bone mineral density compares to their peers.

End of Paper